

How Well Does AI-Generated Feedback Work? Intrinsic and Extrinsic Evaluation across more than 20,000 EFL Essay Drafts

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Abstract. This study examines feedback in English as a Foreign Language (EFL) writing contexts, focusing on written corrective feedback (WCF). Large language models (LLMs) can provide WCF at scale, but aligning them with pedagogical best practices remains an ongoing challenge. WCF meeting criteria like factuality or relevance may still be unsuitable for learning contexts, highlighting the need for extrinsic evaluation based on the learner’s perspective. We deployed WCF systems in a university-level EFL class with nearly 2,000 students, collecting over 20,000 drafts. We evaluated the generated WCF from two perspectives: intrinsic evaluation by experienced English teachers using a rubric, and extrinsic evaluation via student feedback and engagement metrics. Results revealed low alignment between teacher expert ratings and student feedback. These findings suggest that traditional expert evaluation alone may not fully capture WCF’s usability or helpfulness from the learner’s perspective, highlighting the importance of learner-centered evaluation frameworks for AI-based applications in language education.

Keywords: Computer-Assisted Language Learning · Written Corrective Feedback · Writing Assistance · Learning Analytics.

1 Introduction

Learning a second language (L2) is an iterative, cumulative process, during which errors naturally occur. In English as a Foreign Language (EFL) writing education, written corrective feedback (WCF) is often provided to help learners notice

and correct these errors, increasing their linguistic accuracy [1]. However, providing meaningful WCF requires significant time and labor of teachers. Meanwhile, large language models (LLMs) can generate WCF at scale, but evaluating the educational value of this generated text remains a significant challenge [9].

In this study, we explore the evaluation of WCF in EFL writing across two contrasting dimensions: intrinsic aspects (quality as assessed by teachers) and extrinsic aspects (students’ interactions with WCF and perceptions of it). We report on an ongoing large-scale deployment of WCF systems in a university-level EFL class with 1,999 students, analyzing system performance and student interactions across more than 20,000 essay drafts. By comparing teacher and student based metrics, we investigate whether expert-approved WCF aligns with what learners find usable or helpful. We investigate the following research questions:

- **RQ1: Engagement.** Do students use the WCF system? What patterns of engagement emerge across proficiency levels?
- **RQ2: Teacher Perceptions.** How do English teachers perceive the WCF?
- **RQ3: Student Perceptions.** How do students perceive the WCF?
- **RQ4: Evaluation Alignment.** Do teacher perceptions and student perceptions align as measures of WCF quality?

2 Setting and Methods

This study takes place within a fully online English for General Academic Purposes (EGAP) class, a mandatory course for second-year undergraduate students who scored less than 550 points on a TOEFL ITP test.⁶

All students are citizens of Japan and speak Japanese as a first language. The gender distribution is reported as 71.4% male, 28.6% female, though some error may exist due to the lack of any additional categories. The majority of students (n=1701) were of the CEFR framework [3] level B1 based on their TOEFL ITP scores. A smaller number were A2 (n=59), B2 (n=135), or C1 (n=4). One hundred students with missing proficiency scores are excluded from analysis.

The class includes seven graded writing assignments (200–250 words) submitted to a homework website, with up to three drafts per essay allowed. For each submission, students are shown a rule-based automated score that incorporates complexity, accuracy, and fluency (CAF) measures [7]. After submitting a draft, students can access a “Feedback” screen. A screenshot can be seen in Figure 1.

The WCF consists of (i) a highlight that contains the error, and (ii) text that explains the error and what the learner should do. This is divided into two parts, an “explanation” and a “suggestion,” following the approach in Coyne et al. [4]. In our interface, we present these separately as “What’s wrong” and “What to do.” Since the WCF text communicates to the learner about their writing, it constitutes “metalinguistic” feedback [5]. To apply the WCF, students must edit the document themselves, as there is no automatic correction functionality.

⁶ <https://www.ets.org/toefl/itp.html>

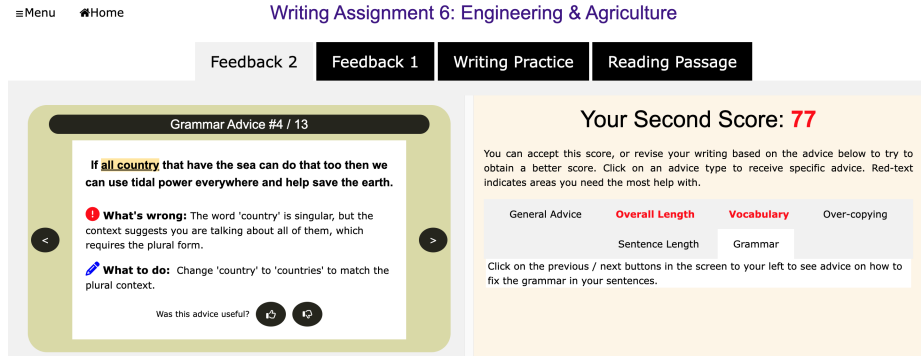


Fig. 1. The writing feedback interface, including WCF with like and dislike buttons. The visual interface for WCF is identical for all systems used in our experiments.

We use three WCF systems, all of which involve the use of an LLM. We use the GPT-4o [10] model `gpt-4o-2024-11-20`. All systems begin with an identical grammatical error correction (GEC) step in which the LLM corrects the draft text and passes “sentence pairs” (original and corrected) to subsequent steps.

The **Template System** is a baseline that does not use LLMs to generate WCF. Instead, WCF is constructed through string operations using text templates. These are selected by comparing the sentence pair using ERRANT [2], which aligns the two strings and produces “edits” annotated with error types, and filling in templates associated with each error type. This system is inspired by ALLECS [11], with the majority of the templates adapted from theirs.

The **Simple LLM System** has the LLM analyze the sentence pair and generate WCF based on the differences between them. We adapt the “keyword-free” prompt from Coyne et al. [4], modifying it so the LLM handles multiple differences between a sentence pair rather than only one. Additionally, the LLM autonomously identifies and highlights errors rather than relying on oracle spans.

The **Atomic Edit System** adds a step to extract and classify edits, forcing one WCF comment for each. This is based on the approach in Song et al. [12], in which “rough edits” are extracted using diffliB and refined into “atomic edits.” We use ERRANT to extract the rough edits, as it is more linguistically informed than diffliB. For error classification, we adopt the typology described in Coyne et al. [4], placing all tags, with up to two examples each, into the prompt.

For system assignment, we adopt a counterbalanced repeated measures design [8] where each student is assigned a different WCF system for each essay assignment, alternating equally between them during the course. The user interface is identical for all systems, whether the WCF is extracted from LLM outputs or built from templates and ERRANT spans. We hypothesize that differences in student rating behavior would thus reflect the quality of the underlying system.

We collect the text of all draft submissions and all generated WCF comments. When students check the WCF, we log whether each comment was viewed. We also incorporate a simple student feedback mechanism in the form of thumbs-

up (“like”) and thumbs-down (“dislike”) buttons, allowing analysis of student engagement and impressions. The dislike button permits a text report.

In all, we collected 20,255 drafts from 1,899 participating students after dropping fragmentary data. Statistics for the full dataset can be seen in Table 1.

Table 1. Dataset Details. All values exclude 100 students with fragmentary data.

Number of drafts	20,255	Total WCF count	144,790
Number of students	1,899	Viewed WCF count	19,470
Number of unique essays	11,545	Drafts with viewed WCF	2,927
Essays completed per student	6.08	Views per draft (excluding zeros)	6.65
Drafts per essay	1.75	Student feedback ratings	597

3 Experiments

3.1 Teacher Evaluation of WCF (RQ2)

To address **RQ2**, we hired four teachers to manually evaluate the generated WCF. Each had at least nine years of EFL teaching experience. We adopt the rubric and task guidelines from Coyne et al. [4], except that raters see fully-corrected sentences rather than isolated corrections. We also add “highlight quality” ratings on a scale of 1-5, and allow raters to flag corrections that are invalid in the first place. The criteria can be seen in the results report in Table 3.

As our systems generated almost 150,000 WCF comments, we conducted the rating task on a sampled subset. We set a target of 600 WCF comments, from which we selected 581 based on our sampling procedures. As we analyze both interaction patterns and system performance, we collect two complementary sets. The “Engagement” set (n=295) comprises WCF that received student feedback via the like or dislike buttons or was viewed, capturing high-engagement contexts and enabling comparison with student perceptions (**RQ4**). This consists of 100 comments with student feedback and up to 50 viewed comments drawn from each of the four CEFR levels. As C1 only had 45 viewed comments, the sampling stops at 295. The “General” set (n=286) is a balanced sample across systems and CEFR levels, with inverse square root weighting by error type to mitigate the dominance of common errors, enabling a more robust evaluation of overall system performance (**RQ2**). Up to 25 WCF comments were selected per stratum. As the C1-Simple System stratum had only 11 comments, sampling stops at 286.

4 Results

4.1 Student Engagement and Perceptions (RQ1, RQ3)

Students can be broadly grouped into those who engaged with WCF and those who ignored it, as 49.76% never accessed the grammar feedback tab. Among

Table 2. Student feedback on WCF comments via the like and dislike buttons.

System	Amount	Sentiment		Dislike Reasons	
	Count (FB Rate)	Likes	Dislikes	Wrong	Highlight
Template	261 (3.46%)	235 (90.0%)	26 (10.0%)	12	2
Simple	168 (2.83%)	160 (95.2%)	8 (4.8%)	3	0
Atomic	159 (2.65%)	148 (93.1%)	11 (6.9%)	5	0

Table 3. Manual rating results for each WCF system. Inter-annotator agreement (α) is calculated on the full data. “Invalid corrections” originate in the shared GEC step.

WCF Rating Criterion	IAA	General Set			Engagement Set		
	(α)	Tmp.	Smpl.	Atm.	Tmp.	Smpl.	Atm.
Correction is valid	0.70	0.92	0.98	0.94	0.97	0.97	0.96
Relevant to error	0.30	0.99	0.99	1.00	0.99	0.97	1.00
Factually correct	0.65	0.92	0.96	0.92	0.96	0.96	0.95
Has what’s wrong & why	0.61	0.28	0.99	0.98	0.41	0.98	0.99
Has what to do	-0.004	0.99	1.00	1.00	0.99	0.99	1.00
Is Comprehensible	0.04	0.98	0.91	0.92	0.98	0.90	0.97
Has out-of-scope ($\downarrow$)	0.22	0.01	0.006	0.005	0.00	0.017	0.013
Has direct correction	0.40	0.84	1.00	1.00	0.84	0.98	1.00
Highlight Quality	0.25	4.54	4.38	4.64	4.65	4.41	4.65
Overall Quality	0.45	3.14	4.51	4.36	3.46	4.45	4.57

those who engaged, the average number of WCF comments viewed was 20.4, with some heavy users viewing 10-20 per draft. The average nonzero views increase with CEFR level among A2 (3.29), B1 (6.57), and B2 (7.98). For C1, it was 5.75.

Likes and dislikes were provided by 259 unique students, of which 71 gave more than one (median of 3). Ten heavy users gave a median of 18.5 responses. Details can be seen in Table 2. We calculate a student feedback rate (FB Rate) by dividing view count by feedback count. This is approximately 3% overall. The majority of student feedback ratings (92.35%) are likes. All text reports referred to the WCF’s factual accuracy or highlight quality, as shown in Table 2.

4.2 Teacher Perceptions (RQ2)

Teacher rating task results can be seen in Table 3. We use Krippendorff’s Alpha to calculate inter-annotator agreement (IAA), finding some limited agreement for correction validity, factuality, and whether the WCF explains what is wrong and why, but low agreement for other criteria. For “out of scope” and “has what to do,” raters marked these as insufficient only 0-2 times per batch, and a lack of overlap on these was penalized severely. Agreement for “Comprehensible” increased with the CEFR level displayed, with -0.18 for A2, 0.17 for B1, and 0.39 for B2. It is null for C1, as all raters selected “Yes” for all such cases.

The two LLM-based systems were rated similarly. The Simple system was rated slightly higher, but a Mann-Whitney U test does not deem it significant ($p=0.628$). Both outperformed the Template system. The Atomic Edit system performed best on highlight quality, which was found significant ($p<0.01$).

4.3 Alignment Between Evaluation Types (RQ4)

We compare the teacher rating results to student feedback on the Engagement set. For the “Overall Quality” 1-5 rating, we take the mean of both ratings per WCF comment. While a small difference is observed for WCF with likes (4.12), dislikes (3.83), and viewed WCF without student feedback (4.04), it was not deemed significant by a Mann-Whitney U Test ($p=0.0625$). Chi-squared tests on binary criteria did not find significant relationships to student feedback.

5 Discussion and Conclusion

With mean ratings between 4 and 5, the LLM-generated WCF was largely approved of by the EFL teachers, and students were more likely to assign likes than dislikes. However, the lack of significant overlap between teacher and student metrics on a per-comment basis suggests that these capture fundamentally different constructs, and that expert rubric-based ratings alone may not fully capture WCF’s usability or helpfulness from the learner’s perspective.

Student perceptions may be primarily based on clarity and immediate usability, but highly usable WCF may not necessarily be optimal for long-term learning, or even free from subtle inaccuracies that expert raters would detect and penalize. Meanwhile, pedagogically rich WCF may be cognitively or linguistically demanding in ways that impede usability, especially for WCF in L2.

Some of the low inter-annotator agreement scores may reflect genuine pedagogical disagreement rather than noise, a pattern noted in prior work on subjective pedagogical judgments [6], and supported here by cases where, during rating task reviews, raters each confirmed their rating and provided a justification. The WCF task involves interpreting learner writing and making decisions about how to communicate to the original writer, leaving significant room for variation when both providing and evaluating WCF. This remains a significant challenge for achieving pedagogical alignment via expert rubric-based evaluation.

This study examined LLM-generated WCF for academic writing in a large, real-world EFL course, comparing criteria-based teacher quality ratings with student interaction and feedback analytics. We find that these measures capture complementary aspects of WCF effectiveness rather than converging, emphasizing the importance of multi-dimensional evaluation of educational applications.

Limitations Our study had several limitations. There were no paired pre-tests and post-tests, nor experimental groups. The student likes and dislikes are sparse signals that are skewed towards high-engagement students. All students shared the same first language. The gender ratio was skewed towards male students, and we were not permitted to use gender on a per-student basis during analysis.

Ethics Students agreed to a consent form permitting all analyses in this study. The participating department handled all personal information and only shared anonymized IDs with the authors. Use of the OpenAI API was approved by the participating department. We did not share essay data with any third party.

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